Explainability and Trustworthiness in Retrieval Augmented Generation (RAG) Systems

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***Abstract*—Retrieval Augmented Generation grounds large language models outputs with obtained information but persistence of few hallucinations remains even after information retrieval. Current RAG applications are nearly completely characterized based on technical features, and human-centered studies of trust and systems' attribution underutilized practical mechanisms to link technical attributes with trust. This paper helps to bridge the existing gap by introducing two approaches to evaluation. Thus, six simple explainability methods belonging to four different classes of mechanisms are implemented in a small-scale RAG prototype based on the SQuAD dataset, and then all of them are evaluated using automated metrics and across a human experiment with 45 participants. In particular, the implemented layer demonstrates very good performance, as the influence scoring value is higher than the retriever's score (76% vs. 68% Recall@1), and it also ranks the correct document at the first position in all queries where the information is retrievable. Moreover, two adapted mechanisms which are independent produce proposals that coincide in 92% of cases. As for the no method is evaluated distinctly better than the rest or from retreived sources and pertubation attribution is rated significantly lower than the rest of the methods in terms of trust($p < 0.05$).**

Index Terms - Retrieval-Augmented Generation, explainable AI, trustworthy AI, large language models, attribution, human-centred evaluation.

I. Introduction

Large Language Models implementation has changed the landscape when it comes to information retrieval tasks. However, their tendency to produce incorrect outputs despite sounding convincing is a problem. RAG has been the main answer to this problem, making it possible to link assembled responses to external data sources[1].

The use of RAG technology has allowed for better fact-checking capabilities, and existing assessments, like CRAG, show that RAG has a clear edge over simple language models [2]. Nevertheless, retrieval cannot guarantee that no inaccurate statements will appear; lots of distortions occur after relevant information has been retrieved, which is a problem in and of itself (Section II). This is the reason why RAG does not eliminate the issue in question; it simply shifts it from the knowledge aspect to the reasoning level.

The issue of dealing with the new trust problem has been met with interest from the XAI community. The development of different RAG techniques from provenance tracing [3] and self reflection [4] to consistency check [5] is worth noting in this regard.

All the above methodologies provide at least partial solutions to the problem of transparency. Nevertheless, they are approached primarily from a technically oriented point of view, like was the explanation produced by the algorithm is true to what the algorithm actually did? The issue of whether these explanations are useful for the people who need to make decisions based on them has not received nearly as much attention. This creates a gap that needs to be filled. The relevant literature on RAG explainability falls into two categories: those using technically solid approaches to attribution and provenance that pass internal validity tests based on consistency within the model, and those focusing on the question of trust without looking at whether the suggested approach has grounded attribution techniques.

Only very few studies analyze the effectiveness of RAG explanations in the two aspects simultaneously, asking not only whether the explanation reflects the workings of the model correctly but also whether the person who was presented with such an explanation trusts the produced output. and Fewer still for small-scale, closed-API RAG, where model internals are unavailable but the need for interpretable explanation is identical.

This paper is designed to solve the problem at hand. We demonstrate and test six easy-to-use explanation techniques for an RAG's Q&A system, specifically document influence scoring, perturbation attribution, provenance tracing, counterfactual explanation, contradiction detection, and consistency checking. The overview of these techniques is provided because they can work without any access to the model's internals. Each technique is assessed using two approaches: automated and human studies. The automated methods are benchmarks for the ranking techniques, whereas the human study tells how successful those explanations are in creating trust in various cases.

II. Related Work

Language models often show tendencies to yield seemingly accurate answers to queries which they do not fully understand [6], leading to the design of Retrieval-Augmented Generation architectures that retrieved evidence guards rather than parametric memory techniques alone [1].The use of RAG logic is preferred than extensive training when it comes to providing advanced knowledge forms [7], [8]. The transition

is notable, but limits exist.CRAG benchmark shows retrieval augmented models enjoying a remarkable superiority over traditional LLMs, but these not totally succeeding with long-tail entities and fast changing information [2], and RAGTruth corpus confirms that models does hallucinations even at the level of retrieval [1]. The grounding process can undergo failures already at earlier stages: as models evaluate document relevance on lexicological criteria, irrelevant documents containing the query terms get accepted as valid proofs [9]. This brought the scholars to understand that hallucination is actually a pipeline problem, which happens because of the complex interaction of retrieval, generation, and validation [10], [11]. Thus, the question of trust was transferred to the place of using the evidence received from retrieval models. [2], [10]

The end-to-end performance of the Retrieval Augmented Generation (RAG) system is extremely dependent on both the retrieval configuration and the context of the retrieved documents [12]. In addition, reliable documents close in relevance can confuse the retrieval process due to semantic mismatch [13]. Surprisingly, research shows that incorporating the irrelevant documents to retrieval output may enhance its accuracy, which indicates that highly relevant retrieval approach may limit the results of generation itself [14]. In this way, the retrieval system is an active factor of the outcome correctness [9], providing the background for picking a semantic retrieval algorithm and considering retrieval performance to be one of the variables influencing the explanation quality.

Reliability leads to complex issues with transparency regarding black-box LLMs do not provide enough reason for trust or accountability [15], which is consistent across machine learning being interspersed with interpretability gaps for methods from accurate but opaque classifiers for botnet detection systems [16]. Human-centred XAI studies argue for explanations directed towards end-users that should respond to the question why and not only what, as well as being empirical [17]. Additionally, it has been stated that users prefer partial breakdown in explanations that suit their reasoning approaches to the comprehensive technical explanation. Thus, in the case of LLMs, token-level post-hoc attribution is not enough, and it is proposed to introduce XAI layer with attention visualization, distinguishing features, and counterfactually determined importance, along with diagnostics [18],[19]. The literature dedicated to trustworthy AI highlights the accuracy of interpretability trade-off along with the idea of transparent design versus post-hoc explanation [20]; defines trust as a complex multi-faceted concept [21],[22]; and develops lifecycle risk frameworks for generative systems [24][25].

The techniques specific to RAG focus directly on the relationship between pieces of evidence and answers. The methods that are faithful and use provenance make it possible for the model to select statements from context and answer only based on given information, thus creating a retraceable artifact [3]. Self-reflection techniques, such as SELF-RAG, help the program learn how to produce tokens of reflection that allow the program to determine if retrieval is necessary and if the retrieved documents are relevant. However, self-reflection techniques depend on training the model, which cannot be done when working with proprietary programs. The consistency methods, such as SelfCheckGPT, identify when lying is present through the process of resampling factual answers remain unchanged while those that are hallucinated diverge [5] which lays the ground for perturbation-based attribution techniques. Closest to the present study, the RAGE framework creates explanations based on retrieval methods for LLMs [26], though they undergo technical validation rather than testing with users.

When considering these types of research, it is evident that two major spheres seldom collide: methods of attribution that are technically sound [3] [5], [26] are validated based on model-centric trust; however, those methods are rarely tested with users. Moreover, research focused on humans and trust [17], [21], [23], which includes the techniques for RAG evaluation being developed [27], looks at perception without taking into consideration attribution methods.

III. Methodology

This work was conducted in a sequence of six stages where (1) the construction of the proposed RAG pipeline was carried out, (2) the implementation of two main explainability methods the document influence scoring technique (Phase 2) and the perturbation-based attribution method (Phase 3) was completed, (3) four additional lightweight methods were developed (Phase 3.5), (4) automated benchmarking was conducted (Phase 4), (5) a human-centered study took place (Phase 5), and (6) the results of both methods were synthesized in a unified evaluation phase of the project (Phase 6). The overall methodology is illustrated in Fig. 1.

A. Design Principles and System Architecture

Three limitations influenced every design choice in this research.The generation model was run using a closed commercial API. This means that no data is available regarding the model's inner workings. Therefore, all six methods used for the explanation are post-hoc, meaning that they rely solely on inputs and outputs of the data given (query, documents, and answer). Methods requiring knowledge of the model's internals (like eg. attention + visualization or SELF-RAG type methods) were not employed due to this limitation.

Low-cost technology. The methods were selected in such a way that mostly no extra computing is required for them to function. four out of six methods (influence scoring,provenance tracing, counterfactual explanation, and contradiction detection) do not require new calls to LLM, as they work with already produced materials. Just perturbation attribution (75 calls) and consistency check (50 calls) require further generation. Hence, the explainability layer can be applied in exact cases targeted in this research.

Dual evaluation by design. Every method is evaluated on two independent criteria automatic benchmarking against proofs and a human study measuring confidence and clarity.

As shown in Fig. 1,The three layer system.at the top the RAG pipeline, followed by the explainability layer(six

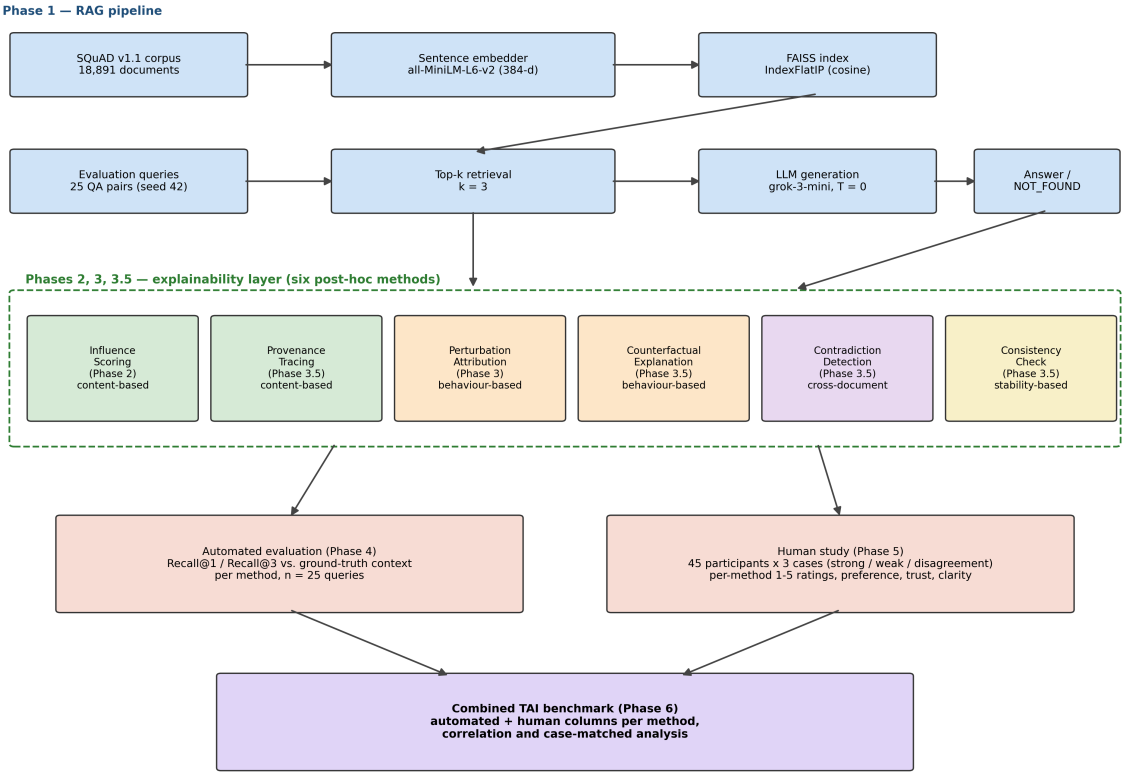


Fig. 1. The pipeline for RAG prototype with dual evaluation (automated/evaluation).

methods), and the evaluation layer (automated and human converging into the combined benchmark).

B. RAG Pipeline Construction

1) Corpus and Evaluation Set: The SQuAD v1.1 training was set as a foundation, the system is constructed [28]. The raw JSON has two types of data created from it. Initially, all paragraphs in the context are compiled and de-duplicated (but the order is preserved) to build up a retrieval corpus consisting of 18,891 paragraphs. Next, all answerable pairs of questions and answers (approximately 87,599 pairs) are extracted. These pairs contain a reference to the paragraph such pairs came from. The existence of this reference to ground-truth context is of great importance, as this is something that allows for validating any of the attribution methods objectively later.

From the QA pool, 25 out of pairs sample is obtained once, utilizing a fixed seed of the random number generator version 42. All samples obtained are stored on the disk. The same pairs make up every stage of the subsequent phases, which guarantees that automated scoring, perturbation results, and human-study cases are performed using the same evaluation set, thus, making it reproducible

2) Semantic Indexing: Each of the 18,891 documents is processed into dense vectors of 384 dimensions via the pre-trained model 'all-MiniLM-L6-v2' [29], with batches of 256 documents processed at once. The use of a semantic embedder over lexical retrieval techniques such as BM25 is due to the tips of the literature review indicating that the models may tend to assess relevance based on lexical overlap rather than real semantic matching phenomena [9], which dense embeddings alleviate but do not fully eliminate.

All embeddings are L2-normalized and indexed using FAISS 'IndexFlatIP'[30]. Once normalized, the inner product becomes equivalent to the cosine similarity, enabling the index to conduct exact cosine-similarity searches for the entire corpus. Exact searches are suitable given the corpus size since retrieval happens within milliseconds while ensuring that the true k-neighbors are not lost. The precise positioning of the index 'IndexFlatIP' and documents is maintained as the index returns their positional identifiers only in the context of the corpus used during the index building.

3) Retrieval Configuration: Retrieval depth was fixed at $k = 3$ on empirical grounds. A pilot depth test over $k \in \{1, 3, 5\}$ showed that when the ground-truth paragraph was retrievable at all, it appeared at rank 1; across the full evaluation set, however, two additional queries recovered their ground-truth document at ranks 2 to 3 (reflected in the gap between Recall@1 and Recall@3 reported in Section IV), while extending to $k = 5$ recovered no further documents in the pilot. $k = 3$ therefore captures all observed coverage benefit while minimizing both the context length passed to

the generator and the cost of the perturbation phase, which requires one regeneration per retrieved document per query. For each query, the retrieval function returns the top-3 documents with their rank, cosine score, and corpus index; the corpus index is retained throughout all later phases so that every attribution result can be traced back to a specific document without text re-matching.

4) Grounded Answer Generation: The 'grok-3-mini' was used for responding via a connection to an API that meets the standards of OpenAI technology. The prompt has been written in such a way so as to guarantee correct grounding: while generating the response, it reiterates that only the data contained in the prompt should be used for creating the response to the question without referencing any other sources, hence allowing the model to say 'NOT_ FOUND' if there is no information available on the subject. The 'NOT_ FOUND' can play two roles: it indicates that the response generated was made with the use of the robotic language of the Machine and it makes possible to produce grounded answers that are fluent – which is the aim of this investigation.

The answers are created at a temperature = 0 and the generation process is capped at 300 tokens. The choice of determinism cannot be considered as a matter of style; instead, it is an obligation, because any change in the outcomes is a result of removing the document through the prism of necessity to produce outputs that are identical each time the method is applied.

C. Explainability Layer

As summarized in Table I, the six techniques intersect four families of mechanisms. Two of the methods are based on content based (in that they focus on content of the retrieved document), the other two are based on behaviour based (they examine how the generated answer varies when influenced), one method is based on cross document comparison (in that it contrasts the current answer with those obtained from the non-primary documents), and one of the methods is based on constructive stability (it observes how robust the same answer is against paraphrasing). The families have been deliberately chosen to represent complementary families – agreement between content-based and behaviour-based methods can be taken as corroborating evidence from independent computations, as is used in Section IV

1) Document Influence Scoring (Phase 2): Document Influence scoring assesses every document found based on a score that ranges from 0 to 1, indicating the extent of its relevancy to the produced answer, by taking into account three factors as follows. $S_{\text{sem}}(q, d)$: this is the semantic matching of the document to the query and is obtained through the computation of the cosine similarity score with the help of the retrieval time techniques. $S_{\text{lex}}(a, d)$: this indicates the answer overlap and tells what percentage of the words contained in the answer given a are found in the document d after excluding all the stop words and converting everything to lowercases. $S_{\text{pres}}(a, d)$: this refers to the presence of the answer and involves a grade of strict matching of the strings, so if the string matched is exactly the same as the given answer the score equals 1, and if there is a match of more than half of the words in the given answer in the searched document the score equals 0.75. If only one word matches, the score equals 0.25. In other cases, the score is 0.

So, the overall score is achieved through the sum of the above-mentioned measures *
I(d) = 0.40 $\cdot S_{\text{sem}}(q, d)$) + 0.35 $\cdot$ $S_{\text{lex}}(a, d)$ + 0.25 $\cdot S_{\text{pres}}(a, d)$.

It should be noted that various configurations of weighting have been studied (default, balanced, more semantic-oriented, and more lexical-oriented). All of them provided the same efficiency of attribution, which means the ranking does not depend on the method used and has allowed retention of the default weighting. Four labels are obtained from rating, namely HIGH INFLUENCE, PARTIAL INFLUENCE, LOW INFLUENCE, and NO INFLUENCE, which were derived by the process of obtaining of their statistics and matching them with the quartiles of the distribution of the actual scores obtained in the evaluation (i.e. Q1 = 0.41, median = 0.85, Q3 = 0.87)

2) Perturbation-Based Attribution (Phase 3): Perturbation attribution gauges the behavioral impact of each respective document: the process consists of eliminating the document, generating a new answer unalteredly and calculating the extent to which the answer has changed. For each query, all three documents are deleted alternately and a new answer is generated (25 queries $\times$ 3 deletions = 75 extra answer generations). The semantic similarity of the initial and perturbed answers, a_0 and a_{-i}, respectively, is calculated using BERTScore F1[31] based on `roberta-large`, which results in sensitivity score.

$$\text{sens}(d_i) = 1 - F_{\text{BERT}}(a_0, a_{-i}), \tag{1}$$

In the case of sens ≈ 0 indicates that a document that was deleted made no difference (the document was no longer important), while sens ≈ 1 shows that the answer was completely adjusted (the document mattered). After running the validation test, three guard rails were created, each addressing a specific problem:

1) **Sentinel handling comes before BERTScore.** According to the validation test results, the encoder assigns an unusually high score (≈ 0.81) to a comparison of the `NOT_FOUND` phrase with an actual answer, unnecessarily reducing sensibility in case of abstention transformations. Thus, the execution of the entire set of rules is carried out without involving BERTScore in all sentinel cases: both `NOT_FOUND` phrases produce a similarity score of 1.0 (no change at all) and only one `NOT_FOUND` phrase leads to a similarity score of 0.0 (the maximum change happened).
2) **Short-answer fallback.** BERTScore cannot be applied reliably to answers with one or two words (which is very common in SQuAD answers), as BERTScore does not work well for such short phrases. Therefore, answers

with ≤ 2 words will be evaluated with the use of string comparison: in case of an exact match, the score will be 1.0, in case of bidirectional substring match, the score will be 0.8, and otherwise, the score will equal 0.0.

3) **Determinism pre-check.** The identity check described in Section III-B4 is a necessary requirement for that stage; it predetermines a possible answer and avoids potential problems with sampling, so that $\text{sens} \approx 0$ means removing the document changed nothing (the document was not load bearing) and $\text{sens} \approx 1$ means the answer changed completely (the document was essential).

The sensitivity score has been transformed into labels like HIGH, MEDIUM, LOW, and NO SENSITIVITY. The boundaries are based on the quartiles of the obtained sensitivity distribution, similarly to what was done in Phase 2. The document that has the highest sensitivity will be taken as the attribution of the query, while the perturbation records will also indicate whether the answer was right in relation to the true answer from high sensitivity on a hallucination.

3) Provenance Tracing (Phase 3.5): The provenance tracing process refines its attribution from document-level to sentence-level based on the faithful reasoning principle of stating where the evidence for a claim is located [3]. The best-matching document will be broken down into individual sentences using punctuation-based segmentation. If the sentence contains the exact answer string, it will be reported with the EXACT label; otherwise, the sentence with the highest overlap of answer content words will be reported as PARTIAL, and if no sentence uses any of the answer's content words, this will be reported as NONE. With respect to abstentions, there is nothing that can be traced, and this is explicitly stated by the process. No regular LLM calls are required for the method.

4) Counterfactual Explanation (Phase 3.5): The counterfactual method uses Phase 3 results and transforms the output into a different form; instead of presenting a number that reflects the degree of sensitivity, it introduces a precise contrast modelled on the sentence "with this source, the system received response X; without this source, the answer was Y". Depending on the statistical measure of dependence, this distinction might take the following forms: COMPLETE(sensitivity $\geq$ 0.8); or when the answer is NOT_FOUND), STRONG ($\geq$ 0.35), PARTIAL ($\geq$ 0.15), NONE. The interesting thing about the development of this method is that it provides a possibility to analyze to what extent the presentation of the information influences the evaluation of trustworthiness.

5) Contradiction Detection (Phase 3.5): In contradiction detection, the retrieved documents except for the primary source are evaluated against the answer, where each document is assigned a stance. The document SUPPORTS the answer if it holds the answer string or at least 60% of its content words; it CONTRADICTS only if it contains negation expressions (e.g., "not ," "never," "incorrect," "disputed") close to the answer terms without accompanying support. The resulting stance from the document is aggregated to yield an overall verdict ("no contradiction," "contradiction detected," or "silent"). There is no availibility for checking any abstentions, where it is reported. Contradiction detection is an intentionally lightweight lexical tool that does not require any LLM calls.Natural-language inference is not carried out, and the inability to recognize any semantic contradictions is noted as a shortcoming, which could possibly be achieved by upgrading to NLI in the future

6) Consistency Checking (Phase 3.5): Consistency check implements the principle that factually grounded answers must not depend on the wording of the question, while fabricated answers differ. AT first, The llm is asked for for each query(temperature=0.7) to generate two paraphrases.After that, the LLM answers two paraphrases of the same question at temperature 0, so the stability score is how many times the answers to the paraphrases coincide with the original answer (bidirectional substring match or joint abstention). The scores are classified into 1.0 STABLE, 0.5 and above MOSTLY STABLE, and less than 0.5 UNSTABLE. Keeping the retrieved set constant is intentional since it allows measuring the generation stability with a fixed set of evidence, although at the cost of losing the end-to-end pipeline stability. The method involves two additional generations per query (50 generations total)

D. Automated Evaluation Protocol

The automated evaluation follows a protocol consisting of two tiers. The retriever, influence scoring, and perturbation attribution each return rankings resulting in a single score only for whether the method gives the location of the true paragraph or not: Recall@1 indicates whether or not the top attribution is the true paragraph, while Recall@3 measures whether or not it is among the top three attributions. Counterfactual explanation is scored the same way while adopting perturbation attribution's score thanks to the dependency levels being equal to the levels of sensitivity scores, of which perturbation attribution's result is dependent on. The last three technologies do not return rankings and cannot be evaluated using attribution recall scores, but rather they are evaluated using signal rates associated with each one of the methods: the exact-trace rate (for provenance tracing), the clean-verdict rate (for contradiction detection), and the stability rate (for consistency checking).

Furthermore,quantitatively accessible which is the case-matched analysis procedure, where verdicts returned by different methods are compared with each other.By comparing the prefix which is 100 characters long in the candidate document as well as the ground-truth paragraph, the results obtained form an accurate comparison because they are correct both under exact string equality as well as under a criterion of word overlap of around 85%. The small size of the evaluation set (which is n= 25) ensures that all the reported recall figures carry with them the Wilson 95% confidence interval which is valid even when n becomes small. Randomly selecting a document from the retrieved documents acts as a random baseline and confirms that the attribution accuracy of the protocol is beyond chance. Moreover, the protocol also computes the cross-method agreement rate between the

perturbation attribution and influence scoring in the protocol, that is the percentage of documents which are defined as the most influential.

E. Human-Centred Evaluation

1) Ethical Approval and Participants: The research study received approval from the School of Computing Ethics Commitee. Introduction of the participants involved the voluntary and anonymous participation. In this way, no identifiable information was required, except the Gmail account for login purposes; however, the email information was not used or recorded. The content for the questionnaire used in the study was extracted from the public SQuAD dataset. The Google Form was filled by forty-five participants and two background questions were asked.

2) Case Selection: Instead of randomly sampling data from survey cases, three examples were identified as part of the Phase 4 analysis in order to represent the totality of the system's behaviour. **Strong case**: an instance in which a query was answered correctly, where the ground-truth document was retrieved at the first document, and where all automated signals agree. **Weak case**: a hallucination: the system generated a plausible-sounding answer that was factually incorrect, whereby the ground-truth answer was never found among the retrieved documents and the retrieved evidence was not supportive. **Disagreement case**: a successful abstention (i.e 'NOT_FOUND')in which the ground-truth document was never retrieved and evidence suggesting the other way was discovered. In these cases, it has been added that a successful explanation should increase trust in the strong case, whereas trust in the weak and disagreement cases should remain low and not be influenced.

3) Survey Format: Each of the three cases used the following format: (i) the question, system's answer, and retrieved sources without explanation – trust rate on a 1-5 scale; (ii) each of the six methods containing an explanation, its description, and what was obtained in that case that received 1-5 score; (iii) a forced choice whether to choose "retrieval only" or an "explainability method;" (iv) binary whether respondent trusts the explainability output, and whether it was clearer than no explanation at all; and (v) if human interpretation was needed (yes/no/no-need more compactness). Therefore, the tool contained 37 fields for every respondent (two background items, 11 items for each of the three cases).

4) Data Quality and Statistical Analysis: The responses are investigated for straight-lining: if a respondent gives the same score to all 21 numeric items (21 numeric items $-$ 7 scores $\times$ 3 scenarios) straight-lining is shown and all the initial analyses must be repeated excluding straight-lining cases. Since the scores represent ordinal data from the Likert scale and considering the small size of the sample, if there is a need to compare ratings of the method under consideration with the ratings from the raw sources, the Wilcoxon signed-rank test will be used instead of any parametric test. The categorical scores (preference, trust, clarity) will be computed as proportions for each case.

F. Benchmark Synthesis

In the last stage of the research, both evaluation aspects are integrated into one unified benchmark, utilizing the semi-automatic assessment category studied in literature on AI evaluation with integrity [25], which involves combining both technical measurements with human evaluations instead of using them independently. To accomplish this, the benchmark table shows the automated Recall@1 and Recall@3 for each method, along with the mean human rating combined for all cases and the ratings for individual cases (strong / weak / disagreement) in the separate columns. The human-only retrieval benchmark is placed in the row corresponding to the retriever, since "primary sources, no interpretation" implies that this situation is same as un-interpreted RAG.

Two analyses are performed on the merged table. The first one is the method-level correlation which is the correlation between the strictly automated rate and the average human ratings. This is described statistically but not pursued, since it is hard to draw any meaningful conclusions with only six different methods available, while the values in the strict column represent a mixture of attribution recall and signal rates. The second analysis focuses on the case-matched validation as the more meaningful type of evaluation. The actual automated data are matched up with the data gathered from the humans in every survey question throughout the analysis. In this way, the gap that could raise questions of validity is closed, as the aggregated automated scores are calculated based on all 25 queries while the human responses cover only 3 chosen cases.

IV. Performance Evaluation

A. Experimental Setup and Reproducibility

The experimentation is done on the Google Colab Platform, utilizing Python Language, sentence-transformers library, faiss library, bert-score library, pandas, and scipy library. The configuration of the full pipeline in this experimentation follows the specification give in section III. The total utilization of LLM comprises 25 initial generations; 75 perturbation generations, 50 paraphrase-generation calls, and finally one paraphrase elicitation call for every query.

The matching mechanism being applied in the last benchmark run is using a prefix of 100 characters. The numbers that we have obtained in our results happen to be the same that have been passed in the actual comparison; the retrieval results in our experiment do occur to be the same from exact string equality as well as from 85% word-overlap point of view (17 out of 25 at the first position; 19 out of 25 within the first three), thus justifying our results.

we used wilcoxon signed rank test because we can compare it paired ordinal for all the statistical data where all test are two sided test. $\alpha = 0.05$ .Moreover , used pearson correlation for discriptive purposes only.

B. Automated Benchmark Results

Table I shows the composite benchmark: attribution recall against the original paragraph for both the retriever and three ranking methods by reporting the method-specific signal rates

TABLE I
COMBINED BENCHMARK. HUMAN RATINGS FROM 1 TO 5 SCALE ($n = 45$);

Method	Automated metric	Strict (@1)	Relaxed (@3)	Human (mean)	Strong	Weak	Disagr.
FAISS Retrieval (baseline)	attribution recall	68%	76%	3.86	3.91	3.84	3.82
Influence Scoring	attribution recall	76%	76%	3.78	4.00	3.76	3.58
Perturbation Attribution	attribution recall	72%	76%	3.46	3.49	3.47	3.42
Counterfactual Explanation†	attribution recall	72%	76%	3.59	3.67	3.58	3.53
Provenance Tracing	signal rate: EXACT / +PARTIAL	64%	76%	3.80	3.84	3.98	3.58
Contradiction Detection	signal rate: clean verdict, answered / overall	72%	96%	3.71	3.93	3.73	3.47
Consistency Check	signal rate: STABLE / ≥MOSTLY STABLE	64%	84%	3.77	4.09	3.76	3.47

(strict / relaxed) used in the study of verdict-based methods, using the two-tier protocol discussed in Section III-D along with the rating of human trust presented in Section IV-C

Attribution performs better than both random performance and its own retrieval rankings. In fact, using random selection results in a Recall@1 of 25.3% (the ground-truth paragraph appears in the top 3 in 19 of 25 cases), and all attribution methods outperform it by a large margin. Thus, influence scoring achieves an impressive Recall@1 of 76%, while retrieval gives a Recall@1 of 68% making the influence scoring correct in all 19 queries where the ground-truth paragraph was retrieved, which is the limit of performance any attribution method can achieve and perturbation attribution performs exactly at the same level as the retriever considering sampling noise. Counterfactual attribution achieves perturbation performance (72% / 76%), since its dependency levels are also based on the same sensitivity scores. Therefore, influence scoring can locate true evidence more reliably than ranking in a plain RAG with no additional inference cost.

The signal rates describe the methods based on verdicts. It was stated in the analysis of the results that in 64% of queries provenance tracing succeeded to capture the right answer word for word from the first document (76% in case of partial matches); when analyzing contradictions found no contradiction in 96% of the queries (with one exception); the consistent checking approach defined 64% of answers as completely stable (84% in part stable) despite paraphrasing. These signal rates indicate the ability to apply the strongest signal of each method, not the exactness of responses, because the most reliable verdict is analyzed in table 3 equipped with related examples.

Uncertainty and consistency of internal approaches. In the case when the number of queries was equal to 25 Wilson interval confidence estimates were high (68% $\rightarrow$ [48.4%, 82.8%]; 76% $\rightarrow$ [56.6%, 88.5%]), thus variations within about $\pm$17 percentage points were directional; the only differences that could be noticed were average baseline and the gap between retrieval and influence.Moreover , Influence scoring and perturbation attribution both of them identified the same most influential document on 23 of 25 queries (92%), despite sharing no computation.

C. Human Study Results

Literal sources fail to offer indication of retrieval failure. The baseline of retrieval alone was estimated at figures 3.91, 3.84 and 3.82, respectively, in strong, weak and disagreement scenarios which remained chronicle, taking into consideration the fact that the truth document was retrieved in only the strong case. Presented with the response as well as raw retrieved passages, research participants failed to differentiate success from failure resulting from the presence of unexplained evidence which does not explain itself otherwise known as the principle of the described study.

Judgement aggregated is accurate. The results for the three categorical metrics decline critically in reliability the preference for the superiority of the explainability addition over pure retrieval reduces from 93% $\rightarrow$ 82% $\rightarrow$ 67%, similar levels of decline are seen in the rate of trust rs from 96% $\rightarrow$ 87% $\rightarrow$ 71% as well as in the clarity rates from 87% $\rightarrow$ 73% $\rightarrow$ 73% respectively. Hence, the values of aggregated judgement perfectly correspond to the actual reliability of cases.

Judgements for each method show dissimilar outcome. The research cover of Wilcoxon method in rating every procedure by cases is provided in Table 2. No method received an overall estimate higher than the base line, meaning that the only one showing significance for the grid is perturbation attribution which was rated below than the raw source in both strong ($p = 0.037$) and weak ($p = 0.035$) cases and showing borderline significance in the disagreement case ($p = 0.079$). The average rating of this approach is the least among its comparable techniques.

Stated vs. revealed preference; compactness. The results are in true contrast: between 67% and 93% of the participants state that they prefer an add-on explaining; however, none of the methods in any of the measures surpass the control, with one of them being well below control the participants agree with the idea of explaining as a concept but end up evaluating all of the testing methods at or below control levels. In response to a question that was not asked, 11%, 18%, and 4% of the respondents (strong/weak/disagreement) said that there was more compactness required from explaining and that length of explaining as such acts as a friction point in the process of evaluating.

Sensitivity to response quality. Sensitivity to quality of answers given. After removing the 6 flagged straight-lining respondents ($n = 39$), all conclusions remain valid, because the perturbation effect remains significant, level differences are at most 0.09, and classification measures have differences of up to 4%.

TABLE II
WILCOXON SIGNED-RANK p-VALUES, ARROWS GIVE THE DIRECTION OF THE METHOD'S MEAN RELATIVE TO THE BASELINE; $*$ MARKS $p < 0.05$.

Method	Strong	Weak	Disagreement
Influence Scoring	0.453 ↑	0.559 ↓	0.182 ↓
Perturbation Attribution	0.037* ↓	0.035* ↓	0.079 ↓
Provenance Tracing	0.674 ↓	0.390 ↑	0.263 ↓
Counterfactual Explanation	0.185 ↓	0.160 ↓	0.152 ↓
Contradiction Detection	0.775 ↑	0.565 ↓	0.254 ↓
Consistency Check	0.206 ↑	0.703 ↓	0.071 ↓

TABLE III
CASE-MATCHED AUTOMATED OUTPUTS AND HUMAN RATINGS FOR THE THREE CASES.

Method	Strong (automated)	Weak (automated)	Disagreement (automated)	Human S / W / D
FAISS Retrieval	ground truth at rank 1	ground truth not retrieved	ground truth not retrieved	3.91 / 3.84 / 3.82
Influence Scoring	0.86 (HIGH)	0.29 (LOW)	0.23 (NO)	4.00 / 3.76 / 3.58
Perturbation Attribution	1.00 (HIGH)	0.14 (LOW)†	1.00 (HIGH)	3.49 / 3.47 / 3.42
Provenance Tracing	EXACT	PARTIAL	NOT FOUND	3.84 / 3.98 / 3.58
Counterfactual Explanation	COMPLETE	COMPLETE	COMPLETE	3.67 / 3.58 / 3.53
Contradiction Detection	NO CONTRADICTION	SILENT	not applicable	3.93 / 3.73 / 3.47
Consistency Check	STABLE (1.0)	UNSTABLE (0.0)	MOSTLY STABLE (0.5)	4.09 / 3.76 / 3.47

D. Discussion

Why is it that influence scoring tracks reliability well but perturbation does not? This is because influence scoring is tied to the quality of the retrieval process as well as the effect of the document on the answer, as calculating the retrieval cosine constitutes 40% of the final score. In this sense, it has been calibrated using the factors that are needed to determine whether the answer is relevant or not. In contrast to this, the removal test does not rely on relevance whatsoever: it only measures dependence, and the disagreement experiment demonstrates how it may happen that the answers drawn from a relied-upon document have no actual relevance.

Why perturbation was deemed to be the least trusted. Beyond not being informative enough to serve as a reliability signal, perturbation claims that the answer will completely change with the removal of the document. A probable interpretation of the considerable negative ratings could be that the framing implies instability the answer is one document away from being different and the participants interpret this correctly as a reason for less trust rather than more. Evidence indicating that counterfactual elaboration same underlying computation presented in a narrative is rated higher in every instance (3.59 vs. 3.46 mean), although it is not statistically significant, supports this interpretation. Presentation format itself seems to influence the trust cost of an identical signal.

Why the clean-verdict rate of contradiction detection is that high. Its 96% is partially justified by favourability of the base rate: in the contexts derived from SQuAD, real cross-document contradictions aren't frequent and thus as a rule a conservative heuristic that makes errors rarely does it wrongly. As shown in Table III it is the only method whose evaluations and reception by humans go in accordance with each other exactly right across the trust spectrum.

Method-level association Using the six methods, the benchmark synthesis looked at the Pearson correlation between the strict automated attribution rate and the average human score ($r = -0.46$; $r = +0.01$ for the relaxed rate). Since the strict column mixes signal rates and attribution recall, and from six points one cannot draw an inference, nothing can be concluded from the data provided. The only interpretable evidence for the divergence is the contrast at the level of methods: perturbation attribution, which is the method that is verified the most from a technical point of view (true attribution recall and 92% agreement with an independent method), is the only one being estimated below unexplained sources, on the contrary, the method of provenance tracing, which has the strictest signal rate joined lowest with the consistent checking, has received the highest mean score (3.80).

What this means for the approach.Overall, the findings support a layered conclusion. Attribution layer performs the intended function, i.e. influence scoring corresponds the actual evidence to the limits of retrieval technology, achieving agreement of two independent methods at the same time in 92% of instances and with four of the six methods introducing no inference cost. The need for attribution technology on the part of participants is also evident, as they do prefer attribution technology over plain retrieval. However, no mechanism measured here can be proved to increase trust in comparison to well-made presentation of raw sources it is worth mentioning that the most carefully validated technology proved to be the least appealing to people. Hence, the gap which this research aimed to detect between the technical fidelity and human-centred appropriateness of the technology is present but inverted for the most technically validated technology; it is necessary to use both evaluative techniques to see the gap, as each used separately would give clear and misleading result.

V. CONCLUSION

The paper addresses a specific gap in retrieval-augmented generation: explanation methods have almost been evaluated on technical, model-based criteria, while their influence on

human normal trust is mostly ignored. In order to measure the true performance of a lightweight closed-API RAG prototype (a corpus of 18,891 SQuAD documents indexed with a FAISS index under strict grounding), we carry out an evaluation of six explanation methods from four mechanism families using two independent measures: an automated benchmark carried out on 25 pre-set queries and a human study involving 45 participants on three different sets of use cases, with the system showing varying reliability levels the case of an adequately grounded response, an apparently confident hallucination, and a valid refusal to answer disguising the failure of a retrieval task.

From a technical point of view, the layer is effective. The influence score is higher than the system's retrieval score (76% against 68% Recall@1), keeping the ground-truth document first among all queries for which it is possible to retrieve it; perturbation attribution matches the retriever within noise (72%); both methods compute independently and agree on the most influencing document in 92% of queries; and four methods out of six do not incur any inference costs. From a human point of view, the picture is inverted. When raw sources are shown, participants fail to distinguish successes from errors in retrieval, their judged results being calibrated well as a group, but there is no individual method rated above the un-explained sources. The method with the highest technical validation, perturbation attribution, received lower ratings in two out of three occasions. Case-matched analysis reveals that removal-based signals indicate only whether the pieces of evidence are strong enough to support the conclusion but does not allow to determine whether the answer is correct or wrong therefore, the best-grounded solutions get a maximum result not differing from that of which are not.

The results are limited by scale (25 different inquiries conducted with the Wilson interval of about ±17 points), a study with 45 individuals, one set of data and generator, and the restrictions related to the methodology used: the implementation of the lexical contradiction rule, a stability check depending on generation rather than analyzing stability, and one computation used in the two approaches.Ultimately, the initial plan specified two techniques and six were presented; a carefully planned, review-based method described as such took its place; moreover, no self-reflection family method was employed here due to the closed-API constraint, as approaches based on reflection tokens such as SELF-RAG [4] call for training access that is not available in this setting.

As Future work,The contradiction heuristic can also be upgraded to an NLI classifier, while the consistency check has to span across multiple paraphrases. Further investigation into the role of explanation framing and length is called for, given how differently the same computation was perceived in two presentations, trust is earned in human terms.

GenAI Usage Disclosure

Generative AI (GenAI) tools were utilized sparingly and in an assistive capacity during the development of this work. When the authors hit technical dead ends, GenAI was used to help debug and resolve coding issues during implementation; all generated code suggestions were reviewed, adapted and validated by the authors before use. In Section IV we also used GenAI to help polish the grammar and improve the presentation and formatting of tables. The research results were not generated by GenAI. The experimental evaluation was not conducted with GenAI. The conclusions from the results were not derived using GenAI.

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